

## MAIN PAPER

# PKBOIN-12: A Bayesian Optimal Interval Phase I/II Design Incorporating Pharmacokinetics Outcomes to Find the Optimal Biological Dose

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## ABSTRACT

Immunotherapies and targeted therapies have gained popularity due to their promising therapeutic effects across multiple treatment areas. The focus of early phase dose-finding clinical trials has shifted from finding the maximum tolerated dose (MTD) to identifying the optimal biological dose (OBD), which aims to balance the toxicity and efficacy outcomes, thus optimizing the risk–benefit trade-off. These trials often collect multiple pharmacokinetics (PK) outcomes to assess drug exposure, which has shown correlations with toxicity and efficacy outcomes but has not been utilized in the current dose-finding designs for OBD selection. Moreover, PK outcomes are usually available within days after initial treatment, much faster than toxicity and efficacy outcomes. To bridge this gap, we introduce the innovative model-assisted PKBOIN-12 design, which enhances BOIN12 by integrating PK information into both the dose-finding algorithm and the final OBD determination process. We further extend PKBOIN-12 to TITE-PKBOIN-12 to address the challenges of late-onset toxicity and efficacy outcomes. Simulation results demonstrate that PKBOIN-12 more effectively identifies the OBD and allocates a greater number of patients to it than BOIN12. Additionally, PKBOIN-12 decreases the probability of selecting inefficacious doses as the OBD by excluding those with low drug exposure. Comprehensive simulation studies and sensitivity analysis confirm the robustness of both PKBOIN-12 and TITE-PKBOIN-12 in various scenarios.

## 1 | Introduction

The 2000s marked the rise of molecularly targeted therapies and immunotherapies, such as imatinib (Gleevec) and trastuzumab (Herceptin), ipilimumab (Yervoy), and pembrolizumab (Keytruda). These medications specifically target and destroy cancer cells by locking onto unique molecular alterations predominantly found in those cells. An increasing number of targeted therapies are now standard treatments for many cancers. For example, immune checkpoint inhibitors have gained widespread application in the treatment of various cancer types,

including melanoma, lung cancer, kidney cancer, bladder cancer, and lymphomas. With the trend of current oncology drug development shifting from traditional chemotherapies to immunotherapies and molecularly targeted therapies, the objective of early phase clinical trials also shifted from determining the maximum tolerated dose (MTD) to identifying the optimal biological dose (OBD), which aims to optimize the toxicity–efficacy trade-off by accounting for both the toxicity and efficacy outcomes. For instance, the chimeric antigen receptor (CAR) T therapies require a balance of boosting the immune system to battle against cancer while avoiding over-stimulation. Unlike

traditional cytotoxic agents in chemotherapies, where efficacy and toxicity are generally assumed to increase with dose, novel agents for targeted therapies may not exhibit a monotonic pattern in the dose–response relationship. Conventional clinical dose-finding trial designs using parametric approaches become more difficult since there are no constraints on the dose–response curve.

The current existing dose-finding designs can be categorized into rule-based, model-assisted, and model-based designs. The 3+3 design is the most commonly used rule-based design because of its simplicity and ease of implementation [1]. Commonly used model-based dose-finding designs for MTD estimation include CRM [2], BLRM [3], EWOC [4], and EWOC-NETS, which is an extension of EWOC to improve trial efficiency and reduce loss of information [5]. Model-based dose-finding designs aimed at OBD detection include Eff-Tox [6], the CP-logistic design [7], the L-logistic design [8], and the B-dynamic design [9]. On the other hand, some popular model-assisted dose-finding designs include mTPI [10], mTPI-2 [11], and BOIN [12, 13]. To optimize the toxicity–efficacy trade-off, an increasing number of model-assisted designs have been developed, such as BOIN-ET [14], BOIN12 [15], STEIN [16], TEPI [11], TEPI-2 [17], PRINTE [18], and uTPI [19]. Among all these designs, BOIN-12 is the only model-assisted dose-finding design to identify the OBD that has received FDA approval for use in early phase clinical trials. For instance, the BOIN12 design was employed in an open-label, non-randomized, single-arm, Phase I/II study of enhanced CD33 CAR T-cells in subjects with relapsed or refractory acute myeloid leukemia to determine the OBD (NCT04835519), in which 25 patients would be enrolled to explore two preset dose levels (dose 1:  $5 \times 10^5 (\pm 20\%)$  CAR T-cells/kg; and dose 2:  $1 \times 10^6 (\pm 20\%)$  CAR T-cells/kg). Another implementation of BOIN12 occurred in a first-in-human (FIH), single-center, open-label, non-randomized, single-arm, Phase I trial aimed at assessing the safety and tolerability of CD5 CAR T-cells in patients with relapsed or refractory T-cell acute lymphoblastic leukemia (NCT05032599). This trial planned to enroll at least 18 subjects to explore two different dose levels (dose 1:  $1 \times 10^6 (\pm 20\%)$  CAR T-cells/kg; and dose 2:  $2 \times 10^6 (\pm 20\%)$  CAR T-cells/kg).

Rapid patient enrollment and delayed onset of toxicity and efficacy outcomes present additional challenges for certain non-cytotoxic anticancer agents, as these factors can slow the assessment process. When the accrual rate is high, some participants may not have finished their outcome evaluations by the time interim decisions need to be made, leading to delays in dose allocation for incoming patient cohorts. Similar problems can occur when toxicity and/or efficacy have a delayed onset, necessitating a longer observation period to accurately assess outcomes. To shorten the trial duration, many model-based or model-assisted designs with time-to-event (TITE) outcomes have been developed, such as TITE-CRM [20], TITE-BOIN [21], TITE-keyboard [22], TITE-EffTox [23], TITE-BOIN12 [24], and TITE-BOIN-ET [25] designs. Based on extensive simulations, these TITE extensions of the dose-finding designs can make real-time decisions with pending outcomes. They can not only reduce trial duration, but also maintain comparable performance in MTD or OBD selection accuracy and patient allocation compared to original dose-finding designs.

Another important objective in early phase dose-finding clinical trials is to evaluate the pharmacokinetics (PK) exposure of the target drug. PK outcomes in clinical trials are measurements that describe the disposition of a drug within the body over a period of time, focusing on the processes of absorption, distribution, metabolism, and excretion. For example, the area under the curve (AUC) measures the total exposure to the drug over time, and the maximum concentration (C<sub>max</sub>) implies the maximum plasma concentration of a drug after administration. PK information can be used to assess drug intervention in humans. Exposure–safety and exposure–efficacy analyses frequently form part of the objectives in early phase clinical trial designs. The PK outcomes can serve as valuable indicators for evaluating various efficacy outcomes of the investigational agent, such as overall response, immune response, and overall survival [26, 27]. Moreover, evidence suggests that subjects with dose-limiting toxicities (DLT) have a statistically significantly larger AUC value compared with subjects with no severe toxicities [28]. This indicates that incorporating PK outcomes into early phase dose-finding clinical trials can improve the accuracy of OBD selection. However, in many trials, the dose-finding process and PK analyses are often conducted separately [29].

There are some dose-finding trial designs in the early phases that have incorporated PK evaluations. Most of these designs are model-based trial designs. Piantadosi and Liu [30] enhanced CRM by integrating PK data as a covariate into a parametric dose–response logit model (PKCOV). Patterson et al. [31] and Whitehead et al. [32] used a hierarchical PK–toxicity model in a cross-over trial setting. Takeda et al. extended the CRM method by incorporating an adjustment based on the latest AUC assessment instead of the standardized dose [33]. These designs mainly focused on pinpointing the MTD based strictly on toxicity outcomes and did not include efficacy outcomes to select the optimal dose considering the benefit–risk balance. Furthermore, these model-based designs that incorporate PK assessments can be complex for researchers without advanced statistical expertise. In addition, these designs are not suitable for clinical trials with late-onset outcomes. To our knowledge, the PK outcomes have not been used in an early phase trial design for the selection of OBD, especially for the late-onset efficacy and toxicity outcomes. Takeda et al. [27] proposed a model-assisted design that made use of all toxicity, efficacy, and PK information. However, their design focused on dose optimization and still does not address the challenge of late-onset outcomes.

In this article, we propose the PKBOIN-12 design, a novel model-assisted dose-finding design that integrates the toxicity, efficacy, and PK outcomes to select the optimal dose. The PKBOIN-12 design extends BOIN12 by incorporating PK data to improve the performance of OBD selection. As a model-assisted design, PKBOIN-12 does not require complex model fitting in real-time during the trial. It can directly use the rank-based desirability score (RDS) table from the BOIN12 framework [15], making it straightforward to implement in actual clinical trials with additional PK data. BOIN12 eliminates a futile dose level only if there is a high posterior probability that its efficacy rate is below the predefined minimum rate during the trial. This limitation causes BOIN12 can select

a futile dose as the OBD, as these futile doses often cannot be eliminated due to a small sample size. Our design considers a positive correlation between the PK outcome with both efficacy and toxicity outcomes. It is reasonable to assume that at low dose levels, certain PK outcomes, such as AUC, would be low, which in turn might be associated with low efficacy rates. By incorporating PK data, PKBOIN-12 can exclude such futile dose levels with unacceptable PK outcomes when determining the OBD, thereby increasing the probability of identifying the optimal dose. For the specific challenge posed by the late-onset efficacy and toxicity outcomes, we have also developed a TITE version of this design, the TITE-PKBOIN-12 design. One advantage of including PK outcomes is the rapidity of PK assessments, which are much quicker to obtain than efficacy and toxicity results. Therefore, we only need to impute the pending toxicity and efficacy outcomes in TITE-PKBOIN-12. Our proposed methods can be used in a wide range of newly emerging dose-finding studies for molecularly targeted therapies. For example, they can be applied to a Phase I dose escalation study which collected safety, efficacy, and PK outcomes of SH-1028 in EGFR T790M-positive advanced non-small cell lung cancer (NSCLC) patients (NCT03603262) [34].

The remainder of this manuscript is organized as follows. In Section 2, we demonstrate the methodology and design algorithms of our proposed PKBOIN-12 and TITE-PKBOIN-12 designs. Section 3 presents comprehensive simulation studies and sensitivity analyses to evaluate the operating characteristics of our proposed designs. In Section 4, we introduce the software developed to implement our proposed designs for practical users. We close with a brief conclusion in Section 5.

## 2 | Methodology

### 2.1 | Optimal Dose Definition

Consider an early phase clinical trial with a total of  $D$  doses. The trial collects a binary toxicity outcome  $Y_T$  and a binary efficacy outcome  $Y_E$ , where  $Y_T = 1$  indicates a DLT and  $Y_E = 1$  represents an efficacy response during the trial. Define  $p_T$  and  $q_E$  as the maximum acceptable toxicity probability and the minimum acceptable efficacy probability, respectively. The true probabilities of toxicity and efficacy for the dose level  $d$  are defined as  $p_d$  and  $q_d$  respectively,  $d = 1, \dots, D$ .  $p_d$  is assumed to increase strictly with the dose level, that is,  $p_1 < \dots < p_D$ . This assumption might not hold for the efficacy outcomes, especially under some newly emerging therapies, such as immuno-oncology therapies and molecularly targeted therapies. Let  $n_{d,T}$  represent the number of patients who experience a DLT and  $n_{d,E}$  denote the number of patients who show an efficacy response at dose level  $d$ . The outcomes based on toxicity and efficacy can be categorized into four types:  $O_1 = (Y_T = 0, Y_E = 1)$ ;  $O_2 = (Y_T = 0, Y_E = 0)$ ;  $O_3 = (Y_T = 1, Y_E = 1)$ ;  $O_4 = (Y_T = 1, Y_E = 0)$ . The patient count for each category  $O_i$  at dose level  $d$  is denoted by  $n_{d,i}$  for  $i = 1, 2, 3, 4$ , satisfying  $n_{d,T} = n_{d,3} + n_{d,4}$  and  $n_{d,E} = n_{d,1} + n_{d,3}$ . Let  $n_d = \sum_{i=1}^4 n_{d,i}$  be the total number of patients assigned to dose level  $d$ . We further define  $\pi_{d,i}$  as the true probability and  $\hat{\pi}_{d,i} = n_{d,i}/n_d$  as the observed probability for each category  $O_i$ . Given the

definitions, it follows that  $p_d = \pi_{d,3} + \pi_{d,4}$ ,  $q_d = \pi_{d,1} + \pi_{d,3}$ ,  $\hat{p}_d = n_{d,T}/n_d$ , and  $\hat{q}_d = n_{d,E}/n_d$ .

Let dose level  $d_{\text{MTD}}$  be the MTD, that is,  $d_{\text{MTD}} = \text{argmax}_d p_d I(p_d \leq p_T)$ . If all dose levels have  $p_d > p_T$ , the MTD does not exist and the trial should be early terminated. Let  $u_i$  be the utility score of  $O_i$  for  $i = 1, 2, 3, 4$  which is pre-specified by the study physician.  $\mathbf{u} = (u_1, u_2, u_3, u_4)$  denotes the utility vector. The best outcome  $O_1$  and the worst outcome  $O_4$  are set at  $u_1 = 100$  and  $u_4 = 0$ , separately. The utility scores for the remaining two outcomes,  $u_2$  and  $u_3$ , fall within the range of  $[0, 100]$ . Given the utility vector  $\mathbf{u}$  and the outcome probability vector  $\boldsymbol{\pi}_d = (\pi_{d,1}, \pi_{d,2}, \pi_{d,3}, \pi_{d,4})$ , the expected utility  $EU_d$  is  $EU_d = \sum_{i=1}^4 u_i \pi_{d,i}$ . A higher value of  $EU_d$  indicates a higher desirability of a dose in the risk-benefit trade-off. In this paper, the OBD is defined as the dose level with the highest  $EU_d$  to optimize the risk-benefit trade-off. When  $u_2 + u_3 = 100$ , we have  $EU_d = u_2(1 - p_d) + u_3 q_d$ . Moreover, if  $u_2 = 0$  and  $u_3 = 100$ , then  $EU_d = 100 q_d$ . In this paper, we set  $(u_2, u_3) = (40, 60)$  as specified in the original BOIN12 paper [15]. However, the summation of  $u_2$  and  $u_3$  is not necessary to be 100.

### 2.2 | Original BOIN12 Design

BOIN12 employs the utility score value to find the OBD and quantify the risk-benefit trade-off [15]. It models the dose utilities using a quasi-beta-binomial model. Lin et al. [15] introduced an RDS for the dosing decision during the trial. Suppose the standardized desirability of the dose level  $d$  as  $u_d = EU_d / 100$ . BOIN12 defines the “quasi-binomial” event  $x_d$  given by  $x_d = 100^{-1} \sum_{i=1}^4 u_i n_{d,i}$ , which can be interpreted as the number of events observed from  $n_d$  patients with a quasi-binomial distribution, that is,  $x_d \sim \text{quasi-binomial}(n_d, u_d)$ . Under the Bayesian framework, BOIN12 assumes that  $u_d$  follows a non-informative uniform prior, specifically Beta(1, 1), which leads to the posterior distribution of  $u_d$  as

$$u_d | n_d, x_d \sim \text{Beta}(1 + x_d, 1 + n_d - x_d).$$

RDS is the rank of the posterior probabilities of  $P(u_d > u_b | n_d, x_d)$  for all possible situations, where the utility benchmark  $u_b = \underline{u} + (100 - \underline{u})/2$  with  $\underline{u} = u_1(1 - p_T)q_E + u_2(1 - p_T)(1 - q_E) + u_3 p_T q_E$ . The dose with a higher RDS is equivalent to having a higher posterior probability  $P(u_d > u_b | n_d, x_d)$ . We presented an example of the RDS table given  $p_T = 0.35$  and  $q_E = 0.25$  as Table S1 in the Supporting Information.

Let  $c = 1$  denote the initial cohort and  $d$  as the lowest or a pre-specified starting dose level. Define  $N^* = 6$  as the pre-specified sample size cutoff. The dose-finding algorithm of BOIN12 follows from Lin et al. [15]:

1. Treat the cohort  $c$  at the dose level  $d$ .
2. Calculate the observed toxicity rate  $\hat{p}_d = n_{d,T}/n_d$  at the current dose level  $d$  and compare it with two constants  $0 < \lambda_1 < \lambda_2 < 1$ , where  $\lambda_1$  and  $\lambda_2$  are the two pre-specified parameters depending on  $p_T$  of BOIN [12].

- if  $\hat{p}_d \geq \lambda_2$ , set  $d'$  as the next lower dose level  $d - 1$ ;
  - if  $\hat{p}_d < \lambda_2$ ,  $n_d \geq 9$ , and the next higher dose level  $d + 1$  has not been used, set  $d'$  to the next higher dose level  $d + 1$ ;
  - if  $\lambda_1 < \hat{p}_d < \lambda_2$ : (i) when  $n_d \geq N^*$ , select one dose  $d'$  from the admissible set  $\{d - 1, d\}$  with larger RDS; (ii) when  $n_d < N^*$ , select one dose  $d'$  from the admissible set  $\{d - 1, d, d + 1\}$  with the largest RDS;
  - if  $\hat{p}_d \leq \lambda_1$ , select one dose  $d'$  from the admissible set  $\{d - 1, d, d + 1\}$  with the largest RDS.
- Set  $c = c + 1$  and update  $d = d'$  based on the dosing decision from Step 2.
  - Repeat Steps 1–3 until the maximum sample size is reached.

BOIN12 selects the OBD after the maximum sample size is reached utilizing the following two steps: (1) determine the MTD by applying isotonic regression to the observed toxicity rates  $\{\hat{p}_d\}_{d=1}^D$  to obtain isotonic estimates  $\{\tilde{p}_d\}_{d=1}^D$  and choose the dose level  $d_{\text{MTD}}^*$  as the MTD such that  $d_{\text{MTD}}^* = \operatorname{argmin}_d |\tilde{p}_d - p_T|$ ; (2) select one dose from  $\{1, \dots, d_{\text{MTD}}^*\}$  as the OBD with the highest estimated utility. To avoid allocating patients to toxic or ineffective doses, BOIN12 considers the following two elimination criteria:

**(Safety)** if  $P(p_d > p_T | \hat{p}_d, n_d) > C_T$ , eliminate dose level  $d$  and all above doses from the dose list;

**(Efficacy)** if  $P(q_d < q_E | \hat{q}_d, n_d) > C_E$ , eliminate dose level  $d$  from the dose list;  $C_T = 0.95$  and  $C_E = 0.9$  are two pre-specified probability cutoffs for toxicity and efficacy, respectively. BOIN12 assumes that  $p_d$  and  $q_d$  follow non-informative beta priors,  $\text{Beta}(1, 1)$ . During the trial, only the doses that fit the two criteria can be assigned to the next cohort of patients. The trial should be terminated without selecting any dose if all dose levels are eliminated.

## 2.3 | Proposed PKBOIN-12 Design

### 2.3.1 | Assumptions of PK Outcome

BOIN12 utilizes toxicity and efficacy information to select the OBD. However, early phase clinical trials also collect PK outcomes. These important PK components provide information on how the drug is absorbed, distributed, metabolized, and eliminated by the body, which is critical to determine the appropriate dosing regimen for the drug and to predict its efficacy and safety at later stages of development. Utilizing these PK outcomes can benefit clinical researchers in identifying OBD, considering that an early phase trial usually has a small sample size with limited information on toxicity and efficacy. Therefore, we propose the PKBOIN-12 design to utilize PK information for OBD identification.

Many continuous PK outcomes are commonly collected in early phase trials, such as AUC, Cmax, Tmax, half-life time

( $t_{1/2}$ ), clearance (CL), and the volume of distribution (Vd). In this paper, we focus on one continuous PK outcome. Let  $r_d$  be the true mean of the PK outcome at dose level  $d$  for analysis, which is presumed to have a positive correlation with both toxicity and efficacy outcomes. Specifically, at the same dose level, patients with a higher PK outcome are more likely to experience both toxic and efficacious outcomes. Similar to the toxicity outcome, the true PK values  $\{r_d\}_{d=1}^D$  are assumed to increase strictly with the dose level, that is,  $0 < r_1 < \dots < r_D$ . Under the Bayesian framework, we assume that each continuous  $r_d$  follows a truncated normal prior left-truncated at 0, truncated  $-N(0, \sigma_0^2; 0, \infty)$  to guarantee  $r_d$  is positive, where  $\sigma_0$  is assumed to be large. By default, we choose  $\sigma_0 = 10000$ . Let  $I_d$  denote the set of all patients assigned to dose level  $d$ . We assume that the individual-level PK value  $r_{d,j} \stackrel{\text{i.i.d.}}{\sim} N(r_d, \sigma_d^2)$ . Therefore, the posterior distribution of  $r_d$  is given by

$$r_d | \hat{r}_d, \sigma_0^2 \sim \text{truncated} - N \left( \frac{n_d \hat{r}_d}{\sigma_d^2 (1/\sigma_0^2 + n_d/\sigma_d^2)}, \frac{1}{1/\sigma_0^2 + n_d/\sigma_d^2}; 0, \infty \right),$$

where  $\hat{r}_d = n_d^{-1} \sum_{j \in I_d} r_{d,j}$  denotes the observed PK sample mean. Therefore, the MAP estimate of  $r_d$  can be approximated by  $\hat{r}_d$  if the small term  $1/\sigma_0^2$  is ignored.

Define  $r_p$  as the target PK value and  $r_1$  as an inefficacious PK value. Similar to the toxicity outcome, consider the following two simple hypotheses:

$$H_{P0}: r_d = r_p \quad \text{vs} \quad H_{P1}: r_d = r_1.$$

By default, we suggest  $r_1 = 0.6r_p$ , which is also recommended by Mu et al. [35] and Takeda et al. [27]. Let  $\zeta_1$  denote the cutoff on the continuous PK outcome, satisfying  $0 \leq \zeta_1 \leq r_p$ . An appropriate  $\zeta_1$  should minimize the posterior probability of incorrect marginal decisions for PK given by

$$P(\text{Incorrect PK decisions} | r_d) = \pi_{P0} P(\hat{r}_d \leq \zeta_1 | H_{P0}) + \pi_{P1} P(\hat{r}_d > \zeta_1 | H_{P1}),$$

where  $\pi_{P0}$  and  $\pi_{P1}$  are the prior probabilities of the two hypotheses of the PK value. Under the assumption of equal non-informative prior probabilities for both hypotheses, set at  $\pi_{P0} = \pi_{P1} = 1/2$ . Therefore,  $P(\text{Incorrect PK decisions} | r_d) = [\Phi(\zeta_1; r_p, \sigma_d^2/n_d) + 1 - \Phi(\zeta_1; r_1, \sigma_d^2/n_d)]/2$ , so that the optimal cutoff value is established as  $\zeta_1 = (r_p + r_1)/2 = 0.8r_p$ . Dose levels with PK outcomes below the threshold  $\zeta_1$  are deemed as potentially ineffective. Adequate exposure to the drug is essential for efficacy, but high PK outcomes alone do not guarantee effectiveness due to the uncertain nature of the dose–response relationship. Additionally, doses with excessively high PK outcomes can pose toxicity risks. Therefore, setting the value of  $r_p$  demands careful consideration, involving thorough discussions with clinicians and PK experts. Researchers should refer to existing PK data from prior clinical trials or preclinical analyses. Unlike the probabilities of toxicity and efficacy, which have ranges between 0 and 1, the ranges for  $r_p$  and  $r_1$  depend on the chosen PK outcome, making it challenging to suggest specific recommended values for these parameters. However, we recommend using a relatively conservative target to mitigate potential safety concerns in the trial.

### 2.3.2 | Dose-Finding Algorithm

The dose allocation algorithm of PKBOIN-12 is based on that of BOIN12. Define  $d_{PK,min}$  as the lowest dose level such that its observed PK sample mean is greater than the cutoff point, that is,

$$d_{PK,min} = \operatorname{argmin}_d \hat{r}_d > \zeta_1.$$

If no dose level achieves  $\hat{r}_d > \zeta_1$ , then  $d_{PK,min}$  does not exist. In PKBOIN-12, we augment the admissible set to include lower dose levels than the current one, provided they have acceptable PK outcomes and less toxicity while maintaining adequate drug exposure. Our proposed design also utilizes the number of quasi-events  $x_d$  and the desirability score for the dosing decisions. PKBOIN-12 inherits all the benefits of using utility scores as seen in BOIN12 and achieves greater flexibility in selecting OBD by integrating PK information. Unlike BOIN12, which limits the admissible set to a maximum of three dose levels, our PKBOIN-12 design expands this to encompass more dose levels with efficacious PK outcomes. For simplicity, we define  $d^*$  as the minimum dose level between dose level  $d - 1$  and  $d_{PK,min}$ , that is,  $d^* = \min\{d - 1, d_{PK,min}\}$ , such that the new constructed admissible set,  $A$ , is a subset of  $\{d^*, \dots, d + 1\}$ . By definition, if  $d_{PK,min}$  does not exist,  $d^* = d - 1$ . However, if  $d_{PK,min}$  exists and is lower than  $d - 1$ , we will expand the admissible set by adding  $\{d^*, \dots, d - 2\}$  to diversify the selection of appropriate dose levels with an efficacious PK value and optimize the desirability for the next cohort of patients.

Let the sample size cutoff  $N^* = 6$ ,  $c = 1$ , and  $d$  as the lowest or a pre-specified dose level. The dose-finding algorithm of the PKBOIN-12 design is shown in Figure 1 and listed below:

1. Treat the cohort  $c$  at the dose level  $d$ .
2. Derive the observed toxicity rate  $\hat{p}_d$ , the observed efficacy rate  $\hat{q}_d$ , and the observed sample mean of the PK value  $\hat{r}_d$  for the current dose  $d$ .
3. If  $\hat{r}_d \leq \zeta_1$ , compare  $\hat{p}_d$  with  $\lambda_1$  and  $\lambda_2$  to determine the following:
  - a. for  $\hat{p}_d \geq \lambda_2$ , set  $d'$  as the next lower dose level  $d - 1$ ;
  - b. for  $\hat{p}_d < \lambda_2$ ,  $n_d \geq 9$ , and the next higher dose level  $d + 1$  has not been used, set  $d'$  to the next higher dose level  $d + 1$ ;
  - c. for  $\lambda_1 < \hat{p}_d < \lambda_2$ , compare  $n_d$  with  $N^*$ . When  $n_d \geq N^*$ , choose one dose  $d'$  from the admissible dose set  $A = \{d - 1, d\}$  with larger RDS; otherwise choose one dose  $d'$  from the admissible dose set  $A = \{d - 1, d, d + 1\}$  with the largest RDS;
  - d. for  $\hat{p}_d \leq \lambda_1$ , choose one dose from the admissible dose set  $A = \{d - 1, d, d + 1\}$  with the largest RDS;
4. If  $\hat{r}_d > \zeta_1$ , compare  $\hat{p}_d$  with  $\lambda_1$  and  $\lambda_2$  to determine the following:
  - a. for  $\hat{p}_d \geq \lambda_2$ , choose one dose  $d'$  from the admissible dose set  $A = \{d^*, \dots, d - 1\}$  with the largest RDS;

- b. for  $\hat{p}_d < \lambda_2$ ,  $n_d \geq 9$ , and the next higher dose level  $d + 1$  has not been used, set  $d'$  to the next higher dose level  $d + 1$ ;
  - c. for  $\lambda_1 < \hat{p}_d < \lambda_2$ , compare  $n_d$  with  $N^*$ . When  $n_d \geq N^*$ , choose one dose  $d'$  from the admissible dose set  $A = \{d^*, \dots, d\}$  with the largest RDS; otherwise choose one dose  $d'$  from the admissible dose set  $A = \{d^*, \dots, d + 1\}$  with the largest RDS;
  - d. for  $\hat{p}_d \leq \lambda_1$ , choose one dose from the admissible dose set  $A = \{d^*, \dots, d + 1\}$  with the largest RDS.
5. Set  $c = c + 1$  and update  $d = d'$  based on the dosing decision of Step 3 or Step 4.
  6. Repeat Steps 1–5 until the maximum sample size is reached.

When  $\hat{r}_d \leq \zeta_1$ , the dosing options in Step 3 are the same as those in Step 2 of BOIN12. However, if  $\hat{r}_d > \zeta_1$ , indicating sufficient PK exposure at the current dose level, the admissible set  $A$  is expanded to include doses  $\{d_{PK,min}, \dots, d - 2\}$ , which have acceptable PK outcomes with lower toxicity probabilities. The design of PKBOIN-12 offers a more flexible set of admissible doses, accommodating various scenarios that can occur during a trial. For instance, if the current dose has an acceptable observed PK outcome and a high observed toxicity rate, that is,  $\hat{r}_d > \zeta_1$  and  $\hat{p}_d \leq \lambda_2$ , PKBOIN-12 chooses the dose with the highest RDS score in the range  $\{d^*, \dots, d - 1\}$ . In a similar situation, BOIN12 would simply move to  $d - 1$ , which means that BOIN12 might need additional cohorts to assign patients to OBD. For example, if the OBD is  $d - 2$ , BOIN12 would require at least two cohorts to start treating new patients with the OBD.

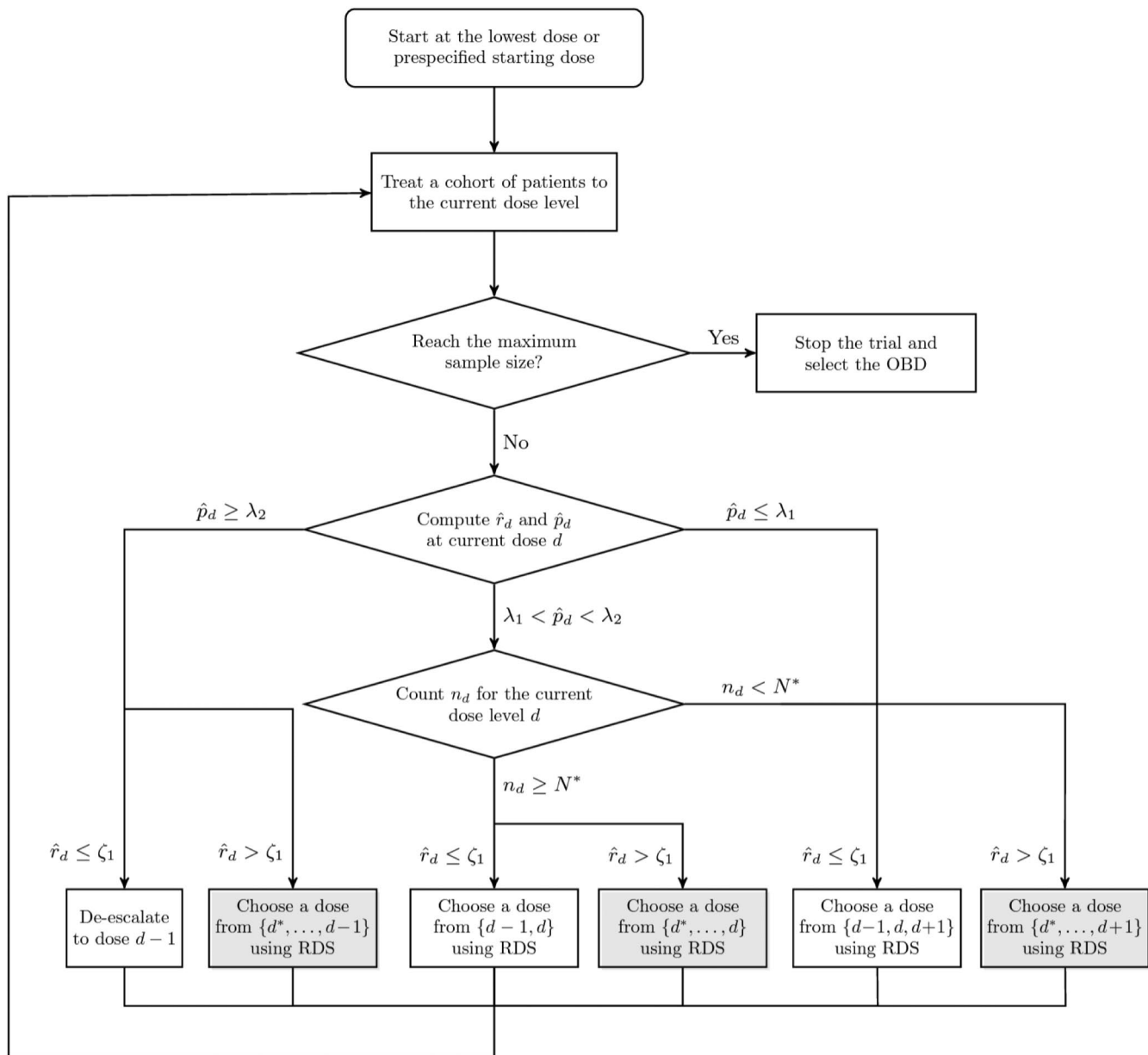
PKBOIN-12 applies the same safety and efficacy criteria as BOIN12 to avoid allocating patients to toxic or inefficacious doses. In addition, we introduce the following elimination criterion based on PK exposure, which removes dose levels with inefficacious PK outcomes from consideration in subsequent dosing decisions:

**(PK)** if  $P(r_d < r_p | \hat{r}_d, n_d) > C_p$  and  $n_d \geq 6$ , eliminate the lowest uneliminated dose level among  $\{1, \dots, d - 1\}$  for  $2 \leq d < D$  and eliminate all the dose levels and terminate the trial for  $d = D$ ,

where  $C_p = 0.95$  denotes the probability cutoff for the continuous PK value. This criterion serves as an additional method for eliminating dose levels that are not effective during the trial, which allows more cohorts to be allocated to the efficacious dose levels. Furthermore, if the highest dose level exhibits a high probability of having a PK exposure below the target PK outcome, all the dose levels will be considered ineffective and the trial will be terminated early.

### 2.3.3 | OBD Selection

Upon completion of the trial, we can estimate the toxicity, efficacy, and PK values for each dose level using the observed data. To ensure a monotonically increasing trend in the toxicity and PK



**FIGURE 1** | The flowchart of the PKBOIN-12 design.

outcomes, we perform isotonic regression models to the observed toxicity probabilities,  $\{\hat{p}_d\}_{d=1}^D$  and PK sample means,  $\{\hat{r}_d\}_{d=1}^D$ , utilizing the pool-adjacent-violators algorithm (PAVA) [36] to obtain the isotonically transformed values  $\{\tilde{p}_d\}_{d=1}^D$  and  $\{\tilde{r}_d\}_{d=1}^D$ . Similar to BOIN12, we use these isotonic estimates of toxicity,  $\{\tilde{p}_d\}_{d=1}^D$ , to identify the MTD. However, unlike BOIN12, which defaults to selecting the first dose level as the lowest in the final admissible set, we integrate isotonic estimates of PK outcomes,  $\{\tilde{r}_d\}_{d=1}^D$ , to determine the lowest dose level for the final admissible set. Specifically, PKBOIN-12 selects the OBD through the following three steps:

1. determine the MTD by applying an isotonic regression to the observed toxicity rates  $\{\hat{p}_d\}_{d=1}^D$  to obtain isotonic estimates  $\{\tilde{p}_d\}_{d=1}^D$  and choose the dose level  $d_{\text{MTD}}^*$  as MTD through  $d_{\text{MTD}}^* = \operatorname{argmin}_d |\tilde{p}_d - p_T|$ ;
2. determine the minimum efficacious PK exposure dose  $d_{\text{PK,min}}^*$  by applying an isotonic regression to the observed PK

sample means  $\{\hat{r}_d\}_{d=1}^D$  to obtain isotonic estimates  $\{\tilde{r}_d\}_{d=1}^D$  such that  $d_{\text{PK,min}}^* = \operatorname{argmin}_d |\tilde{r}_d - r_P|$ ;

3. select one dose from  $\{d_{\text{PK,min}}^*, \dots, d_{\text{MTD}}^*\}$  that has not been eliminated during the trial and has the highest estimated utility value as the OBD.

Unlike the BOIN12 OBD selection procedure, PKBOIN-12 employs a more restricted admissible set for the final OBD selection. This strategy prevents the selection of an ineffective dose level and improves the probability of accurately identifying the OBD. Even in cases where PKBOIN-12 does not correctly identify the OBD, its more refined set of options can still enhance the probability of choosing a dose that is closely aligned with the OBD. This advantage is illustrated in Scenarios 5 and 13 of our simulation studies, where PKBOIN-12's effectiveness is demonstrated through the exclusion of doses that lack efficacy.

## 2.4 | Extension to TITE-PKBOIN-12 Design

Both BOIN12 and PKBOIN-12 utilize complete data from previous cohorts to determine the dosage for the subsequent patient cohort. However, the time required to assess toxicity and efficacy outcomes can span several months, potentially delaying the dosage decision for the next cohort. Some patients' outcomes in the previous cohorts may still be pending when the patients in the next cohort can be enrolled. In contrast, many PK outcomes can be available within a few days after initial treatment. Typically, in clinical trials, the periods for assessing PK outcomes are considerably shorter than those for evaluating toxicity and efficacy outcomes. To make full use of the PK outcomes to shorten the trial duration, we propose the TITE-PKBOIN-12 design as an extension of PKBOIN-12. This extension is similar to TITE-BOIN12, which was introduced as an extension of BOIN12 to deal with the challenges of late-onset outcomes and rapid patient accrual [24].

Let  $A_T$  and  $A_E$  be the toxicity and efficacy assessment time windows, respectively. In TITE-PKBOIN-12, we use individual-level toxicity and efficacy data because each subject has a unique enrollment time, which differs from BOIN12 and PKBOIN-12. Denote  $Y_{j,T}$  as the binary toxicity outcome and  $Y_{j,E}$  as the binary efficacy outcome for the  $j$ th patient. Let  $t$  be the time of the next dose assignment,  $\delta_{j,T}(t)$  and  $\delta_{j,E}(t)$  be the binary variables showing whether  $Y_{j,T}$  and  $Y_{j,E}$  have been determined ( $\delta_{j,T}(t), \delta_{j,E}(t) = 1$ ) or are still pending ( $\delta_{j,T}(t), \delta_{j,E}(t) = 0$ ) at time  $t$ . For example, if  $\delta_{j,T}(t) = 1$ , it indicates that  $Y_{j,T}$  has been observed by time  $t$ ; otherwise,  $Y_{j,T}$  is still pending at time  $t$ . With the approximated likelihood (AL) approach from TITE-BOIN12 [24], the number of quasi-events of dose level  $d$ ,  $x_d$ , can be approximated by  $x_d^*(t)$  at time  $t$  as follows:

$$x_d^*(t) = 100^{-1} \sum_{i=1}^4 u_i n_{d,i}^*(t),$$

where  $n_{d,i}^*(t)$  is an estimate of  $n_{d,i}$  at time  $t$  considering four possible missing patterns: (1) both toxicity and efficacy outcomes are not fully observed; (2) only toxicity outcome is fully observed; (3) only efficacy outcome is fully observed; (4) both toxicity and efficacy outcomes are fully observed. For simplicity, we ignore time point  $t$  in the following notations and formulas. Similar to TITE-BOIN12, we assume that  $Y_{j,T}$  and  $Y_{j,E}$  are independent in TITE-PKBOIN-12. For each outcome  $O_i$ ,  $n_{d,i}^*$  is given by:

$$n_{d,i}^* = \sum_{j \in I_d} \{ I(Y_{j,T} = Y_{i,T}) I(Y_{j,E} = Y_{i,E}) \delta_{j,T} \delta_{j,E} + I(Y_{j,T} = Y_{i,T}) E(Y_{j,E} = Y_{i,E} | \delta_{j,E} = 0) \delta_{j,T} (1 - \delta_{j,E}) \\ + E(Y_{j,T} = Y_{i,T} | \delta_{j,T} = 0) I(Y_{j,E} = Y_{i,E}) (1 - \delta_{j,T}) \delta_{j,E} \\ + E(Y_{j,T} = Y_{i,T} | \delta_{j,T} = 0) E(Y_{j,E} = Y_{i,E} | \delta_{j,E} = 0) (1 - \delta_{j,T}) (1 - \delta_{j,E}) \}, \quad i = 1, 2, 3, 4$$

where  $(Y_{i,T}, Y_{i,E})$  are the toxicity and efficacy outcomes of  $O_i$ . Although the independence assumption is strong, TITE-PKBOIN-12 still has supreme performance in the OBD selection even when this assumption does not hold, as evidenced in the sensitivity analysis presented in Section 3.3. The time to toxicity,  $X_{j,T}$ , and the time to efficacy,  $X_{j,E}$ , are assumed to follow uniform distributions over two intervals,  $[0, A_T]$  and  $[0, A_E]$ , respectively. Based on Yuan et al. [21], the two conditional expectations in the formula for  $n_{d,i}^*$  can be calculated as follows:

$$E(Y_{j,T} = 1 | \delta_{j,T} = 0) = P(Y_{j,T} = 1 | X_{j,T} > t_j) = \frac{\hat{p}_d^* (1 - t_j / A_T)}{1 - \hat{p}_d^* t_j / A_T}, \\ E(Y_{j,E} = 1 | \delta_{j,E} = 0) = P(Y_{j,E} = 1 | X_{j,E} > t_j) = \frac{\hat{q}_d^* (1 - t_j / A_E)}{1 - \hat{q}_d^* t_j / A_E},$$

where  $\hat{p}_d^*$  and  $\hat{q}_d^*$  are estimates of  $p_d$  and  $q_d$  with pending outcomes. Note that  $X_{j,T}$  and  $X_{j,E}$  can be modeled using different distributions, such as the Weibull distribution [21]. In such cases, the results for the two conditional expectations will change. The calculation details of  $\hat{p}_d^*$  and  $\hat{q}_d^*$  are presented in Section A2 of the Supporting Information. With the AL approach, we can estimate  $\{n_{d,i}\}_{i=1}^4$ ,  $\hat{p}_d$ ,  $\hat{q}_d$ , and  $x_d$  by  $\{n_{d,i}^*\}_{i=1}^4$ ,  $\hat{p}_d^*$ ,  $\hat{q}_d^*$ , and  $x_d^*$  for each dose level  $d$ . The detailed dose-finding algorithm of TITE-PKBOIN-12 is presented in Section A3 of the Supporting Information. During the trial, TITE-PKBOIN-12 follows the four criteria below:

**(Safety)** if  $P(p_d > p_T | \hat{p}_d^*, n_d) > C_T$ , eliminate dose level  $d$  and all above doses from the dose list;

**(Efficacy)** if  $P(q_d < q_E | \hat{q}_d^*, n_d) > C_E$ , eliminate dose level  $d$  from the dose list;

**(PK)** if  $P(r_d < r_P | \hat{r}_d, n_d) > C_P$  and  $n_d \geq 6$ , eliminate the lowest uneliminated dose level among  $\{1, \dots, d-1\}$  for  $2 \leq d < D$  and eliminate all the dose levels and terminate the trial for  $d = D$ ;

**(Accrual)** if more than 50% of the patients have pending toxicity or efficacy outcomes at the current dose, suspend the accrual to wait for more data to become available.

We adopt the same pre-specified values of  $\{C_T, C_E, C_P\}$  as those used for PKBOIN-12. In the criteria for safety and efficacy, we substitute  $\hat{p}_d$  and  $\hat{q}_d$  with  $\hat{p}_d^*$  and  $\hat{q}_d^*$ . The PK criterion remains unchanged from what is used in PKBOIN-12, as the PK outcome is presumed to be fully observed by the time of the next dosing decision. These criteria prevent patients from being allocated to toxic or inefficient doses. The accrual criterion is consistent with the one utilized in TITE-BOIN12 [24], which avoids making risky decisions during interim analysis when adequate information is not available. Table 1 presents the de-escalation rule for PKBOIN-12 and TITE-PKBOIN-12, setting  $p_T = 0.35$  with a cohort size of 3 and the above criteria. This demonstrates the feasibility of de-escalating to a lower dose level even in the presence of pending toxicity outcomes. The effective sample size without DLT at time  $t$  is defined as  $m_{d,T}$ . For additional information, refer to Section A2 in the Supporting Information.

After reaching the maximum sample size and all toxicity and efficacy assessments have ended, TITE-PKBOIN-12 selects the final OBD following the same three steps introduced in Section 2.3.3. Similar to PKBOIN-12, TITE-PKBOIN-12 restricts

the final admissible set for OBD selection to only include dose levels with efficacious PK outcomes.

### 3 | Simulation

#### 3.1 | Scenario Configuration

We conducted simulation studies to compare PKBOIN-12 and TITE-PKBOIN-12 with the existing BOIN12 and TITE-BOIN12 designs. We examined  $D = 6$  dose levels, designating dose level 1 as the initial starting point. Each cohort contained three patients, with a total maximum sample size set at 45. The accrual of

patients occurred at a rate of three individuals per month. Both the toxicity and efficacy outcomes were binary outcomes in the simulation studies. The window of toxicity and efficacy were set at  $A_T = 30$  days and  $A_E = 60$  days, respectively. The TITE variables,  $X_{j,T}$  and  $X_{j,E}$ , were assumed to follow two uniform distributions at the intervals  $[0, A_T]$  and  $[0, A_E]$ . We used AUC as the PK outcome for both PKBOIN-12 and TITE-PKBOIN-12. We consistently applied multiple common design parameters to ensure fair and balanced comparisons across all designs. We set the maximum acceptable toxicity probability  $p_T = 0.35$ , the lowest minimum acceptable efficacy probability  $q_E = 0.25$ , and assign the utility scores of  $u_2 = 40$  for the outcome  $O_2$  and  $u_3 = 60$  for outcome  $O_3$ . All designs also shared the same design parameters

**TABLE 1** | De-escalation rule for PKBOIN-12 and TITE-PKBOIN-12 with maximum acceptable toxicity probability  $p_T = 0.35$  and a cohort size of 3.

| $n$ | DLT        | Pending    | De-escalate  | $m_{d,T}$   | $n$ | DLT        | Pending    | De-escalate  | $m_{d,T}$   |
|-----|------------|------------|--------------|-------------|-----|------------|------------|--------------|-------------|
| 3   |            | $\geq 2$   | Pending      |             | 12  | 4          | $3 \sim 6$ | Yes if       | $\leq 5.55$ |
| 3   | 0          | $\leq 1$   | No           |             | 12  | 5          | $1 \sim 6$ | Yes if       | $\leq 6.94$ |
| 3   | 1          | 0          | No           |             | 12  | 6          | $\leq 6$   | Yes          |             |
| 3   | 1          | 1          | Yes if       | $\leq 1.39$ | 12  | $\geq 7$   | $\leq 5$   | Yes and Elim |             |
| 3   | 2          | $\leq 1$   | Yes          |             |     |            |            |              |             |
| 3   | 3          | 0          | Yes and Elim |             | 15  |            | $\geq 8$   | Pending      |             |
|     |            |            |              |             | 15  | $\leq 3$   | $\leq 7$   | No           |             |
| 6   |            | $\geq 4$   | Pending      |             | 15  | 4          | $\leq 5$   | No           |             |
| 6   | $\leq 1$   | $\leq 3$   | No           |             | 15  | 5          | $\leq 3$   | No           |             |
| 6   | 2          | $\leq 1$   | No           |             | 15  | 6          | 0          | No           |             |
| 6   | 2          | $2 \sim 3$ | Yes if       | $\leq 2.77$ | 15  | 4          | $6 \sim 7$ | Yes if       | $\leq 5.55$ |
| 6   | 3          | $\leq 3$   | Yes          |             | 15  | 5          | $4 \sim 7$ | Yes if       | $\leq 6.94$ |
| 6   | 4          | $\leq 2$   | Yes          |             | 15  | 6          | $1 \sim 7$ | Yes if       | $\leq 8.32$ |
| 6   | $\geq 5$   | $\leq 1$   | Yes and Elim |             | 15  | $7 \sim 8$ | $\leq 7$   | Yes          |             |
|     |            |            |              |             | 15  | $\geq 9$   |            | Yes and Elim |             |
| 9   |            | $\geq 5$   | Pending      |             |     |            |            |              |             |
| 9   | $\leq 2$   | $\leq 4$   | No           |             | 18  |            | $\geq 10$  | Pending      |             |
| 9   | 3          | $\leq 1$   | No           |             | 18  | $\leq 3$   | $\leq 9$   | No           |             |
| 9   | 3          | $2 \sim 4$ | Yes if       | $\leq 4.16$ | 18  | 4          | $\leq 8$   | No           |             |
| 9   | $4 \sim 5$ | $\leq 4$   | Yes          |             | 18  | 5          | $\leq 6$   | No           |             |
| 9   | $\geq 6$   | $\leq 3$   | Yes and Elim |             | 18  | 6          | $\leq 3$   | No           |             |
| 9   |            |            |              |             | 18  | 7          | $\leq 1$   | No           |             |
| 12  |            | $\geq 7$   | Pending      |             | 18  | 4          | 9          | Yes if       | $\leq 0.55$ |
| 12  | $\leq 2$   | $\leq 6$   | No           |             | 18  | 5          | $7 \sim 9$ | Yes if       | $\leq 6.94$ |
| 12  | 3          | $\leq 4$   | No           |             | 18  | 6          | 4          | Yes if       | $\leq 8.32$ |
| 12  | 4          | $\leq 2$   | No           |             | 18  | 7          | 2          | Yes if       | $\leq 9.71$ |
| 12  | 5          | 0          | No           |             | 18  | $8 \sim 9$ | $\leq 9$   | Yes          |             |
| 12  | 3          | $5 \sim 6$ | Yes if       | $\leq 4.16$ | 18  | $\geq 10$  |            | Yes and Elim |             |

Note: DLT: Number of patients with DLT, Pending: Number of patients with a pending toxicity outcome,  $m_{d,T}$ : Effective number of patients without DLT, Elim: Eliminate the current and higher dose levels from the trial.

$\lambda_1 = 0.276$ ,  $\lambda_2 = 0.419$ ,  $u_b = 0.705$ , and the RDS tables. The probability cutoff points for toxicity, efficacy, and PK were set at  $C_T = 0.95$ ,  $C_E = 0.9$ , and  $C_P = 0.95$ , respectively. The accrual suspension rule in Section 2.4 was applied to both TITE-BOIN12 and TITE-PKBOIN-12.

In most of the simulation studies of early phase dosing finding publications, the true probabilities of toxicity and efficacy are assumed to be constant for all patients assigned at the same dose level, that is,  $p_{d,j} = p_d$  and  $q_{d,j} = q_d$ , for each  $j \in I_d$ , so that we have  $Y_{j,T} \sim \text{Bernoulli}(p_d)$  and  $Y_{j,E} \sim \text{Bernoulli}(q_d)$ . However, drug exposure can vary among patients at the same dose level. The AUC employed in our simulation studies serves as a measure of the overall drug exposure in the bloodstream over time. A higher AUC value indicates more drug accumulation in the body. The individual-level PK values can have a large variability even at the same dose level. We considered the fact that individual subjects have varying PK outcomes, which can influence their individual toxicity and efficacy responses. The individual-level toxicity and efficacy probabilities,  $p_{d,j}$  and  $q_{d,j}$ , are linked to the actual PK outcome  $r_{d,j}$  [27]. We assumed the coefficient of variation (CV) for the individual-level PK outcome at 25%. To ensure positive values,  $r_{d,j}$  was generated using a truncated normal distribution left-truncated at 0, that is,  $r_{d,j} \sim \text{truncated} - N(r_d, (0.25r_d)^2; 0, \infty)$ . With actual individual-level PK values, the true individual-level toxicity and efficacy probabilities were assumed by

$$p_{d,j} = \min \left\{ p_d \left( 1 + g_p \frac{r_{d,j} - r_d}{r_d} \right), 1 \right\}, \quad \text{and} \quad q_{d,j} = \min \left\{ q_d \left( 1 + g_p \frac{r_{d,j} - r_d}{r_d} \right), 1 \right\},$$

where  $g_p$  is a ratio that measures the relationship of the PK outcome with the true probabilities of toxicity and efficacy at the individual level. A higher  $g_p$  value implies that the variability in the drug exposure plays a more substantial role in affecting toxicity and efficacy outcomes. We set  $g_p = 1$  in our simulations. Sensitivity analyses exploring various values of CV and  $g_p$  are presented in Section 3.3. The toxicity and efficacy outcomes for each patient are generated based on their respective individual-level probabilities. Additional simulations examining the correlation between  $Y_{j,T}$  and  $Y_{j,E}$  are also discussed in Section 3.3.

In our analysis, we considered 14 scenarios. The toxicity and PK curves were assumed to increase monotonically. However, for the efficacy response, we explored four different dose–response relationships:

**Increasing:** response probabilities increase monotonically with dose levels;

**Plateau:** response probabilities increase monotonically until a certain dose level, after which they remain constant;

**Unimodal:** response probabilities increase monotonically until a certain dose level, after which they decrease monotonically;

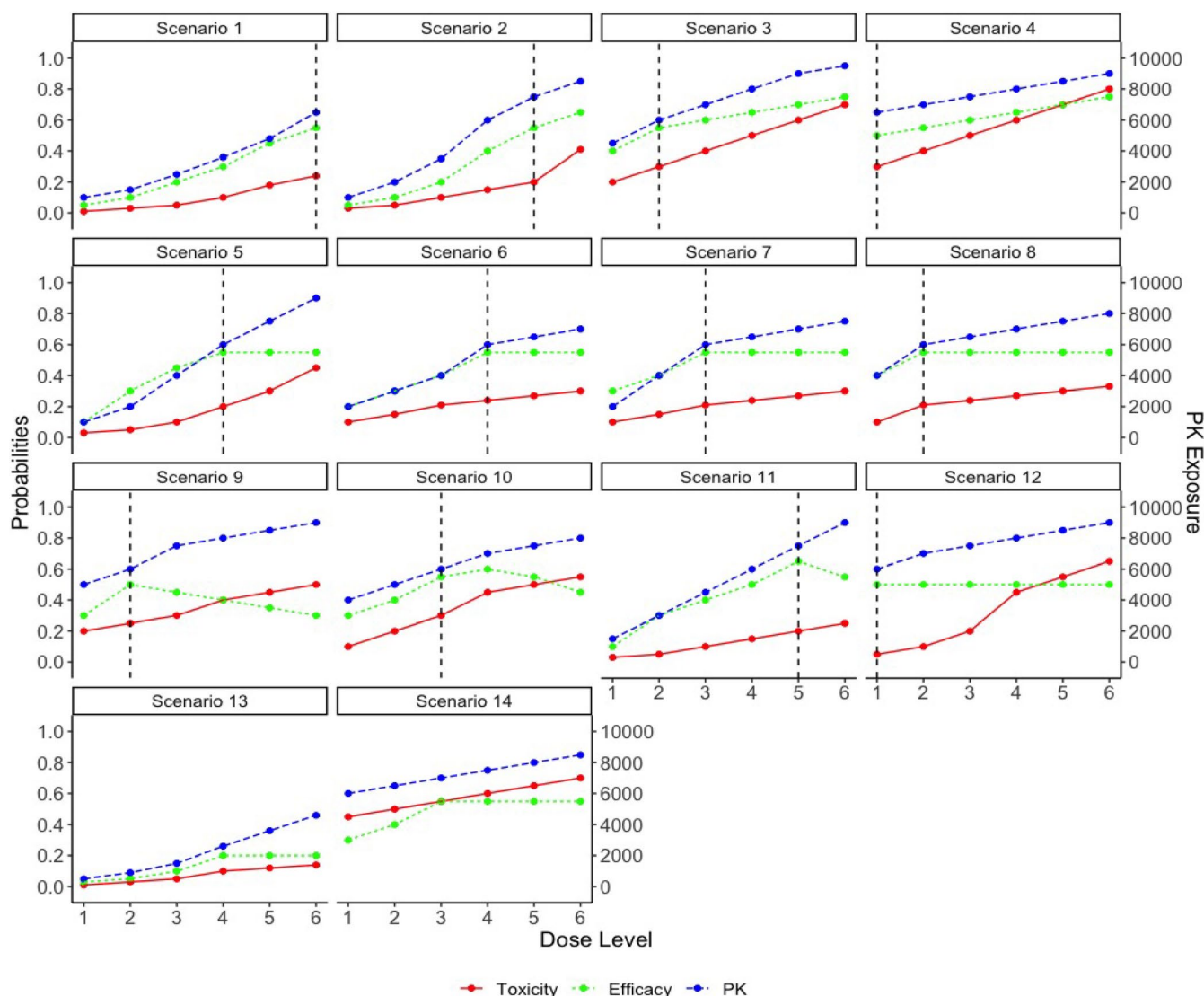
**Constant:** response probabilities have the same value across all dose levels.

The true PK values, along with the probabilities of toxicity and efficacy at each dose level across different scenarios, are shown in Figure 2 and detailed in Table 2. The dose–response relationships and dose levels as the true OBD of all scenarios are presented in Table S2 of the Supporting Information. In Figure 2, we use vertical dashed lines to highlight the true OBD in each scenario, which is also emphasized in bold in Table 3. Scenarios 13 and 14 are two special cases when the true OBD does not exist if all dose levels are too toxic or inefficacious. The target AUC value was set at  $r_p = 6000$ . We conducted 2000 independent replications for each scenario.

### 3.2 | Simulation Results

The simulation results shown in Table 3 include both the proportion of selections as the OBD and the average number of patients assigned to each dose level. Additionally, the table outlines the proportion of early termination and the average duration of the trial in months. Overall, compared to BOIN12 and TITE-BOIN12, PKBOIN-12 and TITE-PKBOIN-12 have higher probabilities of accurately identifying the OBD when the true OBD exists and terminating the trial when the true OBD does not exist. Our proposed designs also allocate more patients to the OBD while allocating much fewer patients to ineffective dose levels compared to BOIN12 and TITE-BOIN12. TITE-PKBOIN-12 has a substantially shorter trial duration than PKBOIN-12. The average trial durations for TITE-PKBOIN-12 and TITE-BOIN12 are nearly the same. However, in Scenario 13, TITE-PKBOIN-12 has a much shorter trial duration than TITE-BOIN12. To prevent redundant comparisons, our comparison results primarily focus on highlighting the differences between PKBOIN-12 and BOIN12.

In the first four scenarios, we considered a monotonically increasing dose–response relationship with different dose levels designated as the true OBD. Scenario 1 set dose level 6 as the true OBD, which was also the only dose level with an effective PK value. PKBOIN-12 had an OBD selection probability of 53.8%, which was 16.9% higher than BOIN12. Dose levels 1, 2, and 3 were ineffective, with total selection probability values of around 13% for BOIN12 and TITE-BOIN12, in contrast to a mere 0% for PKBOIN-12 and TITE-PKBOIN-12. In Scenario 2, the true OBD was dose level 5. PKBOIN-12 had an OBD selection probability of 53.2%, a 3.2% improvement over BOIN12. Both BOIN12 and TITE-BOIN12 continued to have an approximately 10% probability of selecting an ineffective dose level as the OBD. PKBOIN-12 has a 1.5% lower probability of selecting the toxic dose level 6 compared to BOIN12. In Scenario 3, the true OBD was dose level 2. PKBOIN-12 still had a roughly 13.4% greater chance of accurately selecting the OBD compared to BOIN12. In Scenario 4, dose level 1 served as the true OBD. All dose levels possessed effective PK values. All designs showed an approximate 70% probability of correct OBD selection. PKBOIN-12 had a roughly 0.4% higher probability of selecting the OBD compared to BOIN12.



**FIGURE 2** | Curves of true toxicity probabilities, true efficacy probabilities, and true PK values under all scenarios.

In Scenarios 5 through 8, a plateau dose–response relationship was assumed once the target PK exposure level was reached. In Scenario 5, dose level 4 was identified as the OBD, while dose level 5 was the MTD. BOIN12 had an obviously lower probability of accurately selecting the OBD than PKBOIN-12, with 37.7% versus 58.0% respectively. Scenario 5 has the greatest discrepancy in OBD selection probabilities between BOIN12 and PKBOIN-12 when the true OBD existed. PKBOIN-12 exhibited a probability of 19.7% of selecting dose level 3 as OBD. Conversely, BOIN12 had a 12.1% probability of choosing either dose levels 1 or 2, and a 32.6% probability of picking dose level 3 as the OBD. In Scenarios 6, 7, and 8, PKBOIN-12 consistently outperformed BOIN12 by having a greater probability of correctly selecting the OBD and lower probability values of choosing ineffective dose levels.

In Scenarios 9, 10, and 11, a unimodal dose–response relationship was considered. In Scenario 9, dose level 2 served as the OBD and exhibited the highest efficacy probability, all designs had over a 50% probability of selecting the correct OBD. However, PKBOIN-12 had an OBD selection probability of 61.2%, while BOIN12 had a slightly lower probability at 55.6%.

In Scenario 10, PKBOIN-12 displayed a 57.5% probability of OBD selection, marking a substantial 19.1% improvement compared to BOIN12. BOIN12 had a 23.8% probability of selecting ineffective dose level 1, which had a lower toxicity probability but also a much lower efficacy probability compared to the OBD. However, PKBOIN-12 only had merely a 1% probability of selecting that same dose. In Scenario 11, with dose level 5 as the OBD, PKBOIN-12 had a 45.1% probability of OBD selection which was slightly higher than that of BOIN12. PKBOIN-12 displayed a 33.2% probability of selecting the second-best dose, which was 12.0% higher than BOIN12.

Scenario 12 assumed a constant dose–response relationship, making dose level 1 the true OBD. All designs exhibited roughly a 40% probability of correctly identifying the OBD. PKBOIN-12 had a slightly higher probability of selecting the OBD and assigned more patients to the OBD than BOIN12. In Scenario 13, the OBD did not exist because all dose levels were ineffective. PKBOIN-12 and TITE-PKBOIN-12 showed about 85% probability of terminating the trial early without selecting any dose levels. In contrast, both BOIN12 and TITE-BOIN12 had just

**TABLE 2** | True toxicity probabilities, efficacy probabilities, PK, and utility values for each dose level.

| Category | Dose level  |             |             |             |             |             |             |             |             |             |             |      |
|----------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------|
|          | 1           | 2           | 3           | 4           | 5           | 6           | 1           | 2           | 3           | 4           | 5           | 6    |
|          | Scenario 1  |             |             |             |             |             | Scenario 2  |             |             |             |             |      |
| Toxicity | 0.01        | 0.03        | 0.05        | 0.10        | 0.18        | <b>0.24</b> | 0.03        | 0.05        | 0.10        | 0.15        | <b>0.20</b> | 0.41 |
| Efficacy | 0.05        | 0.10        | 0.20        | 0.30        | 0.45        | <b>0.55</b> | 0.05        | 0.10        | 0.20        | 0.40        | <b>0.55</b> | 0.65 |
| PK       | 1000        | 1500        | 2500        | 3600        | 4800        | <b>6500</b> | 1000        | 2000        | 3500        | 6000        | <b>7500</b> | 8500 |
| Utility  | 42.6        | 44.8        | 50.0        | 54.0        | 59.8        | <b>63.4</b> | 41.8        | 44.0        | 48.0        | 58.0        | <b>65.0</b> | 62.6 |
|          | Scenario 3  |             |             |             |             |             | Scenario 4  |             |             |             |             |      |
| Toxicity | 0.20        | <b>0.30</b> | 0.40        | 0.50        | 0.60        | 0.70        | <b>0.30</b> | 0.40        | 0.50        | 0.60        | 0.70        | 0.80 |
| Efficacy | 0.40        | <b>0.55</b> | 0.60        | 0.65        | 0.70        | 0.75        | <b>0.50</b> | 0.55        | 0.60        | 0.65        | 0.70        | 0.75 |
| PK       | 4500        | <b>6000</b> | 7000        | 8000        | 9000        | 9500        | <b>6500</b> | 7000        | 7500        | 8000        | 8500        | 9000 |
| Utility  | 56.0        | <b>61.0</b> | 60.0        | 59.0        | 58.0        | 57.0        | <b>58.0</b> | 57.0        | 56.0        | 55.0        | 54.0        | 53.0 |
|          | Scenario 5  |             |             |             |             |             | Scenario 6  |             |             |             |             |      |
| Toxicity | 0.03        | 0.05        | 0.10        | <b>0.20</b> | 0.30        | 0.45        | 0.10        | 0.15        | 0.21        | <b>0.24</b> | 0.27        | 0.30 |
| Efficacy | 0.10        | 0.30        | 0.45        | <b>0.55</b> | 0.55        | 0.55        | 0.20        | 0.30        | 0.40        | <b>0.55</b> | 0.55        | 0.55 |
| PK       | 1000        | 2000        | 4000        | <b>6000</b> | 7500        | 9000        | 2000        | 3000        | 4000        | <b>6000</b> | 6500        | 7000 |
| Utility  | 44.8        | 56.0        | 63.0        | <b>65.0</b> | 61.0        | 55.0        | 48.0        | 52.0        | 55.6        | <b>63.4</b> | 62.2        | 61.0 |
|          | Scenario 7  |             |             |             |             |             | Scenario 8  |             |             |             |             |      |
| Toxicity | 0.10        | 0.15        | <b>0.21</b> | 0.24        | 0.27        | 0.30        | 0.10        | <b>0.21</b> | 0.24        | 0.27        | 0.30        | 0.33 |
| Efficacy | 0.30        | 0.40        | <b>0.55</b> | 0.55        | 0.55        | 0.55        | 0.40        | <b>0.55</b> | 0.55        | 0.55        | 0.55        | 0.55 |
| PK       | 2000        | 4000        | <b>6000</b> | 6500        | 7000        | 7500        | 4000        | <b>6000</b> | 6500        | 7000        | 7500        | 8000 |
| Utility  | 54.0        | 58.0        | <b>64.6</b> | 63.4        | 62.2        | 61.0        | 60.0        | <b>64.6</b> | 63.4        | 62.2        | 61.0        | 59.8 |
|          | Scenario 9  |             |             |             |             |             | Scenario 10 |             |             |             |             |      |
| Toxicity | 0.20        | <b>0.25</b> | 0.30        | 0.40        | 0.45        | 0.50        | 0.10        | 0.20        | <b>0.30</b> | 0.45        | 0.50        | 0.55 |
| Efficacy | 0.30        | <b>0.50</b> | 0.45        | 0.40        | 0.35        | 0.30        | 0.30        | 0.40        | <b>0.55</b> | 0.60        | 0.55        | 0.45 |
| PK       | 5000        | <b>6500</b> | 7500        | 8000        | 8500        | 9000        | 4000        | 5000        | <b>6000</b> | 7000        | 7500        | 8000 |
| Utility  | 50.0        | <b>60.0</b> | 55.0        | 48.0        | 43.0        | 38.0        | 54.0        | 56.0        | <b>61.0</b> | 58.0        | 53.0        | 45.0 |
|          | Scenario 11 |             |             |             |             |             | Scenario 12 |             |             |             |             |      |
| Toxicity | 0.03        | 0.05        | 0.10        | 0.15        | <b>0.20</b> | 0.25        | <b>0.05</b> | 0.10        | 0.20        | 0.45        | 0.55        | 0.65 |
| Efficacy | 0.10        | 0.30        | 0.40        | 0.50        | <b>0.65</b> | 0.55        | <b>0.50</b> | 0.50        | 0.50        | 0.50        | 0.50        | 0.50 |
| PK       | 1500        | 3000        | 4500        | 6000        | <b>7500</b> | 9000        | <b>6000</b> | 7000        | 7500        | 8000        | 8500        | 9000 |
| Utility  | 44.8        | 56.0        | 60.0        | 64.0        | <b>71.0</b> | 63.0        | <b>68.0</b> | 66.0        | 62.0        | 52.0        | 48.0        | 44.0 |
|          | Scenario 13 |             |             |             |             |             | Scenario 14 |             |             |             |             |      |
| Toxicity | 0.01        | 0.03        | 0.05        | 0.10        | 0.12        | 0.14        | 0.45        | 0.50        | 0.55        | 0.60        | 0.65        | 0.70 |
| Efficacy | 0.03        | 0.05        | 0.10        | 0.20        | 0.20        | 0.20        | 0.30        | 0.40        | 0.55        | 0.55        | 0.55        | 0.55 |
| PK       | 500         | 900         | 1500        | 2600        | 3600        | 4600        | 6000        | 6500        | 7000        | 7500        | 8000        | 8500 |
| Utility  | 41.4        | 41.8        | 44.0        | 48.0        | 47.2        | 46.4        | 40.0        | 44.0        | 51.0        | 49.0        | 47.0        | 45.0 |

merely about 0.3% probability of early termination. Additionally, PKBOIN-12 had an average trial duration of 26.9 months, which was approximately 13 months shorter than that of BOIN12.

PKBOIN-12 enrolled only about 10 cohorts of patients before ending the trial, whereas BOIN12 did not terminate early. Scenario 14 was another scenario without the true OBD because all dose

**TABLE 3** | Simulation results for all designs with  $p_T = 0.35$ ,  $q_E = 0.25$ , including selection probability for each dose level, number of patients assigned to each dose level, and average trial duration.

|                | Selection probability (%) |      |      |      |      |      | Number of assigned patients |      |      |      |      |      | Duration |          |
|----------------|---------------------------|------|------|------|------|------|-----------------------------|------|------|------|------|------|----------|----------|
|                | 1                         | 2    | 3    | 4    | 5    | 6    | ET                          | 1    | 2    | 3    | 4    | 5    | 6        | (Months) |
| Design         | Scenario 1                |      |      |      |      |      |                             |      |      |      |      |      |          |          |
| BOIN12         | 1.7                       | 2.5  | 8.9  | 19.1 | 30.8 | 36.9 | 0.1                         | 3.8  | 4.4  | 5.8  | 8.2  | 11.5 | 11.4     | 38.4     |
| PKBOIN-12      | 0.0                       | 0.0  | 0.1  | 3.3  | 40.9 | 53.8 | 1.9                         | 3.0  | 3.3  | 5.1  | 8.2  | 12.6 | 12.6     | 38.1     |
| TITE-BOIN12    | 1.3                       | 2.5  | 8.6  | 18.3 | 32.0 | 37.2 | 0.1                         | 3.7  | 4.4  | 6.0  | 8.4  | 11.5 | 10.9     | 25.2     |
| TITE-PKBOIN-12 | 0.0                       | 0.0  | 0.0  | 3.2  | 41.9 | 52.8 | 2.0                         | 3.2  | 3.6  | 5.4  | 8.5  | 12.7 | 11.4     | 25.0     |
| Design         | Scenario 2                |      |      |      |      |      |                             |      |      |      |      |      |          |          |
| BOIN12         | 1.6                       | 2.5  | 6.0  | 22.7 | 50.0 | 17.2 | 0.1                         | 3.9  | 4.5  | 5.5  | 9.5  | 14.5 | 7.0      | 37.9     |
| PKBOIN-12      | 0.0                       | 0.0  | 2.4  | 28.5 | 53.2 | 15.7 | 0.3                         | 3.1  | 4.1  | 6.1  | 10.9 | 14.2 | 6.6      | 37.9     |
| TITE-BOIN12    | 1.2                       | 2.4  | 6.3  | 23.8 | 47.9 | 18.2 | 0.1                         | 4.0  | 4.5  | 5.7  | 10.0 | 13.8 | 6.9      | 24.7     |
| TITE-PKBOIN-12 | 0.0                       | 0.0  | 2.5  | 28.1 | 52.0 | 17.1 | 0.3                         | 3.3  | 4.2  | 6.1  | 11.0 | 13.9 | 6.5      | 24.7     |
| Design         | Scenario 3                |      |      |      |      |      |                             |      |      |      |      |      |          |          |
| BOIN12         | 35.9                      | 45.9 | 14.3 | 2.3  | 0.0  | 0.0  | 1.5                         | 17.3 | 17.5 | 7.5  | 1.9  | 0.3  | 0.0      | 36.7     |
| PKBOIN-12      | 18.7                      | 59.2 | 17.0 | 2.4  | 0.2  | 0.0  | 2.4                         | 16.2 | 18.6 | 7.4  | 1.8  | 0.3  | 0.0      | 36.5     |
| TITE-BOIN12    | 35.9                      | 44.3 | 15.3 | 2.5  | 0.2  | 0.0  | 1.7                         | 17.0 | 16.9 | 7.8  | 2.3  | 0.4  | 0.0      | 20.8     |
| TITE-PKBOIN-12 | 19.4                      | 59.2 | 16.1 | 2.8  | 0.2  | 0.0  | 2.2                         | 16.1 | 18.3 | 7.6  | 2.0  | 0.4  | 0.0      | 20.7     |
| Design         | Scenario 4                |      |      |      |      |      |                             |      |      |      |      |      |          |          |
| BOIN12         | 69.8                      | 19.2 | 2.5  | 0.1  | 0.0  | 0.0  | 8.3                         | 27.8 | 10.8 | 3.1  | 0.5  | 0.0  | 0.0      | 34.8     |
| PKBOIN-12      | 70.2                      | 18.2 | 2.6  | 0.2  | 0.0  | 0.0  | 8.7                         | 28.3 | 10.5 | 2.9  | 0.5  | 0.1  | 0.0      | 34.9     |
| TITE-BOIN12    | 69.4                      | 20.6 | 2.8  | 0.5  | 0.0  | 0.0  | 6.6                         | 27.0 | 11.5 | 3.5  | 0.7  | 0.1  | 0.0      | 19.4     |
| TITE-PKBOIN-12 | 70.4                      | 19.7 | 2.8  | 0.2  | 0.0  | 0.0  | 6.8                         | 28.0 | 10.9 | 3.2  | 0.7  | 0.1  | 0.0      | 19.3     |
| Design         | Scenario 5                |      |      |      |      |      |                             |      |      |      |      |      |          |          |
| BOIN12         | 1.0                       | 11.1 | 32.6 | 37.7 | 15.3 | 2.1  | 0.0                         | 3.7  | 7.3  | 12.6 | 12.6 | 6.6  | 2.1      | 37.7     |
| PKBOIN-12      | 0.0                       | 0.0  | 19.7 | 58.0 | 19.2 | 2.9  | 0.1                         | 3.2  | 6.3  | 13.3 | 13.8 | 6.5  | 1.9      | 37.6     |
| TITE-BOIN12    | 0.6                       | 10.9 | 31.7 | 37.9 | 16.2 | 2.7  | 0.0                         | 3.9  | 7.8  | 12.7 | 12.3 | 6.4  | 2.0      | 23.4     |
| TITE-PKBOIN-12 | 0.0                       | 0.0  | 18.4 | 58.2 | 20.3 | 2.8  | 0.1                         | 3.5  | 7.2  | 13.1 | 12.9 | 6.4  | 1.9      | 23.4     |

(Continues)

TABLE 3 | (Continued)

|                | Selection probability (%) |      |      |      |      |     |     | Number of assigned patients |      |      |      |     |     | Duration |
|----------------|---------------------------|------|------|------|------|-----|-----|-----------------------------|------|------|------|-----|-----|----------|
|                | 1                         | 2    | 3    | 4    | 5    | 6   | ET  | 1                           | 2    | 3    | 4    | 5   | 6   | (Months) |
| Design         | Scenario 6                |      |      |      |      |     |     |                             |      |      |      |     |     |          |
| BOIN12         | 7.3                       | 14.4 | 18.7 | 34.0 | 18.6 | 6.7 | 0.3 | 7.1                         | 8.9  | 9.2  | 10.7 | 6.2 | 2.7 | 37.9     |
| PKBOIN-12      | 0.1                       | 1.7  | 17.2 | 47.1 | 21.6 | 9.7 | 2.5 | 4.6                         | 7.9  | 10.7 | 11.9 | 6.3 | 3.0 | 37.4     |
| TITE-BOIN12    | 6.9                       | 14.1 | 17.2 | 35.7 | 17.7 | 8.1 | 0.3 | 7.3                         | 9.0  | 9.2  | 10.6 | 6.0 | 2.8 | 23.4     |
| TITE-PKBOIN-12 | 0.2                       | 1.5  | 16.6 | 46.1 | 23.4 | 9.9 | 2.3 | 5.1                         | 8.0  | 10.8 | 11.5 | 6.3 | 2.8 | 23.4     |
| Design         | Scenario 7                |      |      |      |      |     |     |                             |      |      |      |     |     |          |
| BOIN12         | 9.1                       | 17.0 | 34.8 | 23.2 | 11.3 | 4.5 | 0.1 | 7.9                         | 10.0 | 12.9 | 8.2  | 4.1 | 1.8 | 37.5     |
| PKBOIN-12      | 0.2                       | 10.2 | 45.1 | 26.0 | 12.4 | 4.9 | 1.0 | 6.0                         | 10.8 | 14.3 | 8.1  | 3.9 | 1.6 | 37.3     |
| TITE-BOIN12    | 9.0                       | 18.2 | 35.9 | 20.7 | 10.8 | 5.2 | 0.2 | 8.4                         | 10.7 | 13.0 | 7.6  | 3.6 | 1.7 | 22.6     |
| TITE-PKBOIN-12 | 0.4                       | 10.6 | 46.3 | 25.1 | 11.7 | 5.3 | 0.7 | 6.7                         | 11.1 | 14.1 | 7.9  | 3.5 | 1.5 | 22.5     |
| Design         | Scenario 8                |      |      |      |      |     |     |                             |      |      |      |     |     |          |
| BOIN12         | 20.2                      | 37.9 | 21.4 | 12.2 | 5.6  | 2.6 | 0.2 | 12.0                        | 15.9 | 8.9  | 4.7  | 2.2 | 1.1 | 37.1     |
| PKBOIN-12      | 8.7                       | 47.0 | 25.3 | 12.3 | 4.8  | 1.8 | 0.1 | 11.8                        | 17.0 | 9.3  | 4.4  | 1.7 | 0.7 | 37.1     |
| TITE-BOIN12    | 19.2                      | 37.6 | 22.8 | 12.7 | 5.2  | 2.3 | 0.2 | 12.5                        | 15.8 | 9.0  | 4.7  | 2.1 | 0.9 | 21.6     |
| TITE-PKBOIN-12 | 8.8                       | 48.3 | 24.3 | 12.2 | 4.6  | 1.7 | 0.2 | 12.2                        | 17.1 | 9.2  | 4.2  | 1.6 | 0.6 | 21.5     |
| Design         | Scenario 9                |      |      |      |      |     |     |                             |      |      |      |     |     |          |
| BOIN12         | 18.0                      | 55.6 | 19.7 | 3.8  | 0.9  | 0.1 | 2.0 | 12.3                        | 18.0 | 8.8  | 3.4  | 1.4 | 0.5 | 37.4     |
| PKBOIN-12      | 9.8                       | 61.2 | 20.8 | 4.4  | 0.5  | 0.2 | 3.0 | 12.6                        | 18.8 | 8.4  | 3.0  | 1.0 | 0.4 | 37.3     |
| TITE-BOIN12    | 17.3                      | 54.5 | 20.8 | 4.3  | 0.9  | 0.2 | 1.9 | 12.1                        | 17.9 | 9.0  | 3.5  | 1.4 | 0.5 | 22.1     |
| TITE-PKBOIN-12 | 9.8                       | 62.2 | 20.4 | 4.2  | 0.8  | 0.1 | 2.5 | 12.7                        | 18.7 | 8.5  | 3.1  | 1.0 | 0.3 | 21.9     |
| Design         | Scenario 10               |      |      |      |      |     |     |                             |      |      |      |     |     |          |
| BOIN12         | 23.8                      | 28.2 | 38.4 | 8.8  | 0.5  | 0.0 | 0.3 | 12.2                        | 13.3 | 13.3 | 5.0  | 1.0 | 0.2 | 37.7     |
| PKBOIN-12      | 0.9                       | 25.9 | 57.5 | 12.7 | 1.2  | 0.1 | 1.8 | 8.3                         | 15.1 | 14.9 | 5.2  | 1.0 | 0.2 | 37.3     |
| TITE-BOIN12    | 22.2                      | 28.6 | 38.4 | 9.4  | 1.0  | 0.0 | 0.3 | 11.8                        | 13.3 | 13.2 | 5.3  | 1.1 | 0.2 | 22.1     |
| TITE-PKBOIN-12 | 1.2                       | 25.9 | 57.0 | 12.7 | 1.6  | 0.1 | 1.5 | 8.7                         | 14.9 | 14.5 | 5.3  | 1.2 | 0.2 | 22.0     |

(Continues)

TABLE 3 | (Continued)

|                | Selection probability (%) |      |      |      |             |      |             | Number of assigned patients |      |      |      |             |     | Duration |
|----------------|---------------------------|------|------|------|-------------|------|-------------|-----------------------------|------|------|------|-------------|-----|----------|
|                | 1                         | 2    | 3    | 4    | 5           | 6    | ET          | 1                           | 2    | 3    | 4    | 5           | 6   | (Months) |
| Design         | Scenario 11               |      |      |      |             |      |             |                             |      |      |      |             |     |          |
| BOIN12         | 0.4                       | 10.1 | 15.0 | 21.2 | <b>43.1</b> | 10.2 | 0.0         | 3.7                         | 7.2  | 8.4  | 9.9  | <b>11.6</b> | 4.2 | 37.5     |
| PKBOIN-12      | 0.0                       | 0.0  | 10.4 | 33.2 | <b>45.1</b> | 10.9 | 0.2         | 3.2                         | 6.1  | 9.7  | 10.6 | <b>11.3</b> | 4.1 | 37.5     |
| TITE-BOIN12    | 0.4                       | 9.8  | 15.6 | 22.6 | <b>42.9</b> | 8.8  | 0.0         | 3.9                         | 7.7  | 9.2  | 10.1 | <b>10.6</b> | 3.6 | 23.7     |
| TITE-PKBOIN-12 | 0.0                       | 0.0  | 12.3 | 33.4 | <b>43.6</b> | 10.5 | 0.1         | 3.5                         | 7.0  | 10.3 | 10.4 | <b>10.2</b> | 3.6 | 23.7     |
| Design         | Scenario 12               |      |      |      |             |      |             |                             |      |      |      |             |     |          |
| BOIN12         | <b>41.6</b>               | 37.2 | 19.0 | 1.8  | 0.3         | 0.0  | 0.0         | <b>17.2</b>                 | 14.8 | 9.0  | 3.0  | 0.8         | 0.1 | 37.3     |
| PKBOIN-12      | <b>44.5</b>               | 36.5 | 17.0 | 1.8  | 0.1         | 0.0  | 0.0         | <b>18.6</b>                 | 15.3 | 8.0  | 2.5  | 0.5         | 0.1 | 37.3     |
| TITE-BOIN12    | <b>43.6</b>               | 34.3 | 19.7 | 2.1  | 0.2         | 0.0  | 0.0         | <b>18.1</b>                 | 14.8 | 8.6  | 2.7  | 0.7         | 0.2 | 21.1     |
| TITE-PKBOIN-12 | <b>45.1</b>               | 35.5 | 17.2 | 2.1  | 0.1         | 0.0  | 0.0         | <b>20.1</b>                 | 14.7 | 7.4  | 2.2  | 0.5         | 0.1 | 21.0     |
| Design         | Scenario 13               |      |      |      |             |      |             |                             |      |      |      |             |     |          |
| BOIN12         | 6.9                       | 8.3  | 13.7 | 31.5 | 23.8        | 15.5 | <b>0.3</b>  | 5.2                         | 6.1  | 7.1  | 9.6  | 9.5         | 7.5 | 39.8     |
| PKBOIN-12      | 0.0                       | 0.0  | 0.0  | 0.5  | 4.8         | 10.0 | <b>84.7</b> | 3.0                         | 3.3  | 4.3  | 6.7  | 7.2         | 5.8 | 26.9     |
| TITE-BOIN12    | 5.8                       | 6.5  | 13.9 | 31.8 | 25.2        | 16.8 | <b>0.2</b>  | 5.0                         | 5.8  | 6.9  | 9.7  | 9.8         | 7.9 | 27.5     |
| TITE-PKBOIN-12 | 0.0                       | 0.0  | 0.0  | 0.7  | 4.0         | 10.2 | <b>85.0</b> | 3.1                         | 3.4  | 4.4  | 6.9  | 7.3         | 5.8 | 20.8     |
| Design         | Scenario 14               |      |      |      |             |      |             |                             |      |      |      |             |     |          |
| BOIN12         | 39.0                      | 3.8  | 0.8  | 0.2  | 0.0         | 0.0  | <b>56.3</b> | 22.9                        | 5.5  | 1.8  | 0.4  | 0.0         | 0.0 | 26.3     |
| PKBOIN-12      | 40.5                      | 3.7  | 0.9  | 0.1  | 0.0         | 0.0  | <b>54.8</b> | 23.1                        | 5.5  | 1.8  | 0.3  | 0.0         | 0.0 | 26.5     |
| TITE-BOIN12    | 45.8                      | 3.3  | 0.6  | 0.1  | 0.0         | 0.0  | <b>50.2</b> | 24.3                        | 6.0  | 1.9  | 0.4  | 0.1         | 0.0 | 15.9     |
| TITE-PKBOIN-12 | 44.5                      | 4.6  | 0.9  | 0.1  | 0.0         | 0.0  | <b>49.9</b> | 24.2                        | 6.0  | 2.0  | 0.5  | 0.1         | 0.0 | 15.9     |

levels were toxic. In this scenario, BOIN12 and PKBOIN-12 had a roughly similar probability of early termination, exceeding 50%. However, BOIN12 had a 1.5% higher probability of early termination compared to PKBOIN-12. The average trial duration in both designs was approximately 26.5 months.

Our simulation study revealed that PKBOIN-12 and TITE-PKBOIN-12 can effectively address the issue of a high probability of selecting an ineffective dose level as the OBD during the trial by incorporating PK information. This issue can result in no effective dose levels advancing to the dose-expansion phase or Phase II study, thereby causing the failure of the entire clinical development. In Scenarios 1, 2, 6, 11, and 13, where ineffective dose levels were present, both BOIN12 and TITE-BOIN12 displayed considerably higher probabilities of selecting these dose levels as the OBD. However, the probabilities of selecting toxic doses as the OBD were comparable across all four designs. Based on the simulation outcomes, our TITE-PKBOIN-12 design is most favorable, because of its high probability of accurate OBD selection, low risk of selecting ineffective doses, and shortest trial duration.

### 3.3 | Sensitivity Analysis

We conducted a set of sensitivity analyses to assess the performance and robustness of both PKBOIN-12 and TITE-PKBOIN-12 designs. Firstly, we compared PKBOIN-12 with BOIN12 by varying the  $g_p$  parameter through the values  $\{0, 0.5, 1, 1.5, 2\}$ . If  $g_p = 0$ , the PK outcome had no relationship with the toxicity and efficacy outcomes at the individual level. Table 4 presented the OBD selection probabilities of PKBOIN-12 and BOIN12, considering various  $g_p$  values across all scenarios. PKBOIN-12 consistently outperformed BOIN12 for different  $g_p$  values. Even in the cases where the PK outcome had no impact on toxicity

and efficacy outcomes, that is,  $g_p = 0$ , PKBOIN-12 still had superior performance. This is because PKBOIN-12 can exclude dose levels with ineffective PK outcomes from OBD selection. As  $g_p$  increased, the OBD selection probability for PKBOIN-12 generally increased across the majority of scenarios. In the second sensitivity analysis, we adopted two alternative CV values, 0.1 and 0.4, and presented the simulation results for BOIN12 and PKBOIN-12 in Tables S4 and S5 of the Supporting Information. A high CV indicates substantial variability in individual-level PK outcomes. PKBOIN-12 was observed to consistently outperform BOIN12 in all scenarios with varying CV values.

Additionally, we evaluated the robustness of PKBOIN-12 and TITE-PKBOIN-12 when the toxicity and efficacy outcomes were correlated. We conducted two separate simulation studies given the correlation coefficients between toxicity and efficacy outcomes as  $\rho_{pq} = 0.2$  and  $\rho_{pq} = 0.4$ . The outcomes are presented in Table S6 and Table S7, respectively. Despite the design algorithms of PKBOIN-12 and TITE-PKBOIN-12 operating under the independence assumption between the toxicity and efficacy outcomes, both designs demonstrated robustness across all examined scenarios. When there was a positive correlation between toxicity and efficacy, the OBD selection probabilities for our designs tended to increase slightly in most scenarios. We further examined the performance of two proposed designs under different utility score settings, specifically  $(u_{11}, u_{00}) = (100, 0)$ . This setting focused on identifying the dose level with the highest efficacy probability while keeping the toxicity probability not greater than  $p_T$ . As illustrated in Table S8 of the Supporting Information, both PKBOIN-12 and TITE-PKBOIN-12 maintained similar probabilities for OBD selection across all scenarios. Finally, we assessed the performance of the two designs with longer assessment windows, denoted as  $(A_T, A_E) = \{(45, 90), (90, 120)\}$ . From Tables S9 to

**TABLE 4** | OBD selection probabilities (%) for BOIN12 (B) and PKBOIN-12 (P) across all scenarios given different values of  $g_p$ .

| Scenario | $g_p = 0$ |      | $g_p = 0.5$ |      | $g_p = 1$ |      | $g_p = 1.5$ |      | $g_p = 2$ |      |
|----------|-----------|------|-------------|------|-----------|------|-------------|------|-----------|------|
|          | B         | P    | B           | P    | B         | P    | B           | P    | B         | P    |
| 1        | 37.1      | 51.6 | 37.4        | 51.8 | 36.9      | 53.8 | 38.2        | 56.0 | 37.4      | 57.0 |
| 2        | 49.6      | 51.4 | 45.6        | 52.3 | 50.0      | 53.2 | 51.6        | 51.5 | 50.6      | 52.7 |
| 3        | 46.1      | 58.2 | 44.8        | 58.3 | 45.9      | 59.2 | 46.5        | 57.2 | 46.5      | 58.3 |
| 4        | 69.7      | 70.1 | 69.8        | 69.3 | 69.8      | 70.2 | 69.2        | 70.8 | 70.8      | 70.3 |
| 5        | 40.5      | 56.9 | 38.0        | 56.6 | 37.7      | 58.0 | 37.2        | 58.0 | 41.1      | 61.0 |
| 6        | 32.8      | 44.4 | 31.9        | 43.9 | 34.0      | 47.1 | 34.4        | 46.8 | 34.8      | 47.0 |
| 7        | 35.7      | 44.6 | 36.7        | 44.2 | 34.8      | 45.1 | 35.9        | 45.9 | 34.3      | 47.9 |
| 8        | 37.5      | 47.2 | 35.0        | 44.4 | 37.9      | 47.0 | 36.4        | 45.8 | 37.0      | 45.6 |
| 9        | 55.9      | 59.9 | 55.8        | 61.3 | 55.6      | 61.2 | 56.9        | 61.6 | 57.0      | 62.5 |
| 10       | 38.7      | 57.6 | 38.0        | 55.3 | 38.4      | 57.5 | 38.2        | 59.5 | 38.6      | 57.8 |
| 11       | 43.4      | 44.6 | 44.8        | 45.8 | 43.1      | 45.1 | 44.5        | 46.0 | 42.8      | 43.0 |
| 12       | 42.8      | 42.4 | 41.1        | 44.8 | 41.6      | 44.5 | 43.6        | 46.1 | 41.1      | 45.6 |
| 13       | 0.1       | 84.5 | 0.4         | 85.0 | 0.3       | 84.7 | 0.2         | 85.8 | 0.2       | 86.2 |
| 14       | 57.0      | 56.9 | 55.5        | 57.0 | 56.4      | 54.8 | 55.3        | 55.5 | 56.5      | 59.2 |

Note: The true OBD in Scenarios 13 and 14 do not exist, so the presented probability values represent the probability of early termination.

**S10**, both designs continued to exhibit robust probabilities for OBD selection and patient assignment to OBD with different assessment windows. With longer assessment windows, TITE-PKBOIN-12 demonstrated a greater ability to reduce trial duration. Compared to PKBOIN-12, TITE-PKBOIN-12 reduced the average trial duration by approximately 10.2 months for  $(A_T, A_E) = (30, 60)$ , 14.9 months for  $(A_T, A_E) = (45, 90)$ , and 19.0 months for  $(A_T, A_E) = (90, 120)$ .

## 4 | Software

The R programs for implementing BOIN12 and TITE-BOIN12 are available at [www.trialdesign.org](http://www.trialdesign.org). The programming code for our proposed PKBOIN-12 and TITE-PKBOIN-12 designs is available at <https://github.com/EugeneHao/PKBOIN-12>.

## 5 | Conclusion

Early phase dose-finding clinical trials often collect PK outcomes as they serve as valuable indicators for assessing the initial safety and efficacy of the study drug based on exposure-safety and exposure-efficacy analyses. We developed the innovative PKBOIN-12 design, which integrates PK data with toxicity and efficacy outcomes to identify the optimal dosage. Furthermore, to address the issue of late-onset toxicity and efficacy outcomes, we introduced the TITE-PKBOIN-12 design. Both PKBOIN-12 and TITE-PKBOIN-12 are extensions of BOIN-12 and can employ utility scores and the RDS table for OBD selection, enhancing the risk-benefit trade-off. Similar to TITE-BOIN12, TITE-PKBOIN-12 employs an approximated likelihood method for estimating the unobserved count of quasi-events. Thus, both PKBOIN-12 and TITE-PKBOIN-12 are straightforward to implement in clinical trials, with the additional PK outcome for dosing decisions. Although we utilize AUC as the PK outcome in our simulation studies, it is important to note that AUC is not the only option available. Different PK outcomes exhibit varying distributions, such that they cannot be incorporated into the design with the same design parameters. However, our designs only require the assumption that the continuous PK outcome increases monotonically with the dose level, which is similar to the assumption of the toxicity outcome. The purpose of employing PK outcomes is to minimize the likelihood of assigning patients to doses with insufficient drug exposure during the trial and to eliminate such doses from the admissible set for the final selection of OBD. The PK outcome does not influence the allocation and selection procedure for doses with acceptable drug exposure. Therefore, we believe that PK-related design parameters, such as  $r_p$ , CV, and  $C_p$ , have more impact on the effectiveness of the design compared to the specific PK outcome chosen. Note that our proposed designs may not be suitable for a novel treatment without prior PK information. While practical users can set an extremely low target PK value of  $r_p$  to avoid eliminating acceptable doses during the trial, the benefit of incorporating the PK outcome can be minimal. In such cases, the performance of our designs would be very similar to that of the original BOIN12 and TITE-BOIN12 designs.

In our simulation studies, we explored four different dose-response relationship patterns. Our simulations also considered the variability in individual outcomes due to varying levels of

drug exposure. The simulation results demonstrated that, compared to BOIN12, PKBOIN-12 increased the probability of correctly identifying the OBD when the true OBD existed and the probability of terminating the trial without selecting any dose as the OBD when the true OBD does not exist. PKBOIN-12 had a lower chance of selecting futile dose levels as OBD than BOIN12. Furthermore, PKBOIN-12 was more effective in assigning patients to the optimal dose than BOIN12. These advantages stem from utilizing the PK outcome to expand the admissible set at the dose exploration step and to exclude dose levels with unacceptable PK outcomes during the OBD selection step. Our sensitivity analyses revealed that PKBOIN-12 and TITE-PKBOIN-12 had robust performance. PKBOIN-12 consistently outperformed BOIN12 under varying conditions of individual PK outcome and differing associations between PK outcome and toxicity and efficacy outcomes. Even in the sensitivity analysis where we assumed the PK outcome was uncorrelated with the toxicity and efficacy outcomes, that is,  $g_p = 0$ , PKBOIN-12 still outperformed BOIN12 in terms of OBD selection probability.

Our proposed designs, PKBOIN-12 and TITE-PKBOIN-12, have certain limitations. First, they incorporate only one continuous PK outcome, restricting the ability to use additional collected PK data. If physicians and pharmacometricians would like to include multiple PK outcomes, our current design would not suffice. An extension of our design involves accommodating multiple PK outcomes. This can be achieved by developing a composite PK outcome that combines various PK outcomes using the Mahalanobis distance or a weighted average. The composite PK outcome remains continuous and will increase monotonically if all included PK outcomes increase monotonically with dose level. In that case, our proposed designs can still be applied using this composite PK outcome. Additionally, it is possible to replace some of the continuous PK outcomes with binary or categorical outcomes. Second, our designs focus primarily on binary toxicity and efficacy outcomes. However, like BOIN12 and TITE-BOIN12, our designs can be adapted to accommodate categorical toxicity and efficacy outcomes by constructing a new utility table as the combination of the two categorical variables, with utility values established through discussions with clinicians. Third, our designs do not account for subgroup effects. One possible solution is to stratify patients into subgroups and apply the designs individually to each subgroup. However, this method might lead to an increase in the sample size requirement. Some subgroups may experience slower enrollment rates, which could prolong the duration of the trial. Fourth, our proposed designs consider only the PK outcomes, without incorporating pharmacodynamic (PD) outcomes. Future extensions could integrate PD data, considering the relationships between PK, PD, toxicity, and efficacy. Fifth, in our current designs, we assumed that individual-level PK outcomes follow a normal distribution. While this assumption simplifies the modeling, it may not hold in all cases, as real-world PK data can exhibit skewness or extreme values. If the normality assumption is violated, it could impact the optimal dose selection and the design performance. Future research could explore the robustness of our method under alternative distributional assumptions. Finally, TITE-PKBOIN-12 assumes that the individual-level PK data will be available in a short time. However, in real-world clinical trials, delays in accessing PK data are common due to the extensive data collection and processing procedures. As a result, the

reduction in trial duration promised by TITE-PKBOIN-12 may not be as pronounced in practical clinical trials as indicated by our simulation results.

## Conflicts of Interest

The authors declare no conflicts of interest.

## Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

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### Supporting Information

Additional supporting information can be found online in the Supporting Information section.